occurred leading to the loss of restriction site that can be digested by EcoRI.

The result is that when the DNA molecules are digested by the enzyme Hind III, there is no difference in the DNA fragments separated. However, with the enzyme EcoRI, source A DNA molecules is not digested while source B DNA molecule is digested. This results in a polymorphic pattern of separation.

Random amplified polymorphic DNA (RAPD) is a molecular marker based on PCR amplification. An outline of RAPD is depicted in Figure 12.35. The DNA isolated from the genome is denatured. The template molecules are annealed with primers, and amplified by PCR. Single short oligonucleotide primers (usually a 10-base primer) can be arbitrarily selected and used for the amplification of DNA segments of the genome (which may be distributed throughout the genome). The amplified products are separated on electrophoresis and identified.

Based on the nucleotide alterations in the genome, the polymorphisms of amplified DNA sequences differ which can be identified as bands on gel electrophoresis. Genomic DNA from two different sources often results in different amplification patterns i.e. RAPDs.

Infobits

In 1970s American molecular biologists Allan M. Maxam and Walter Gilbert and English biochemist Frederick Sanger developed some of the first techniques for DNA sequencing. Gilbert and Sanger shared the 1980 Nobel Prize for Chemistry for their work. Dideoxy procedure is one of the procedure used to sequence DNA.

Summary

The fundamental unit of information in living systems is the gene. Genome is the set of all genes and genetic signals of a cell. Gene is expressed through a sequence of events. The central dogma of molecular biology, comprises the three major processes replication, transcription and translation. The genetic message encoded in messenger RNA (mRNA) is translated on the ribosomes into a polypeptide with a particular sequence of amino acids. An RNA base sequence (a set of 3 bases) corresponding to a particular amino acid is called a codon. The genetic code is the set of all codons. The genetic code is a triplet code, and all 64 possible codons carry information of some sort. The code is highly redundant. Polypeptides are synthesized from the amino terminus toward the carboxyl terminus, by adding amino acids one by one to the carboxyl end. An important feature of intiation of polypeptide synthesis in both prokaryotes and eukaryotes is the use of a specific initiating tRNA molecule.

Mutation is the process by which the sequence of base pairs in a DNA molecule is altered. Mutations can be divided into

TN_GOVT_XII_Micro_Biology_CH12.indd 285 04-03-2019 10:51:56